

Friction

Surface friction is, by definition, a force acting parallel to a surface of contact. The amount depends on particular surfaces, type of contact, and specific materials.

Friction may be a nuisance to designers for some machines but it is a necessity for every day life. Walking, sitting, writing, eating, etc. The use of greases and oil between bearing surfaces reduces friction by providing a layer of molecules that are free to slide past each other.

For objects of ordinary size and speed moving through a fluid (air & liquid), the frictional force, or drag force, is proportional to the cross sectional area of the object and to the square of its speed.

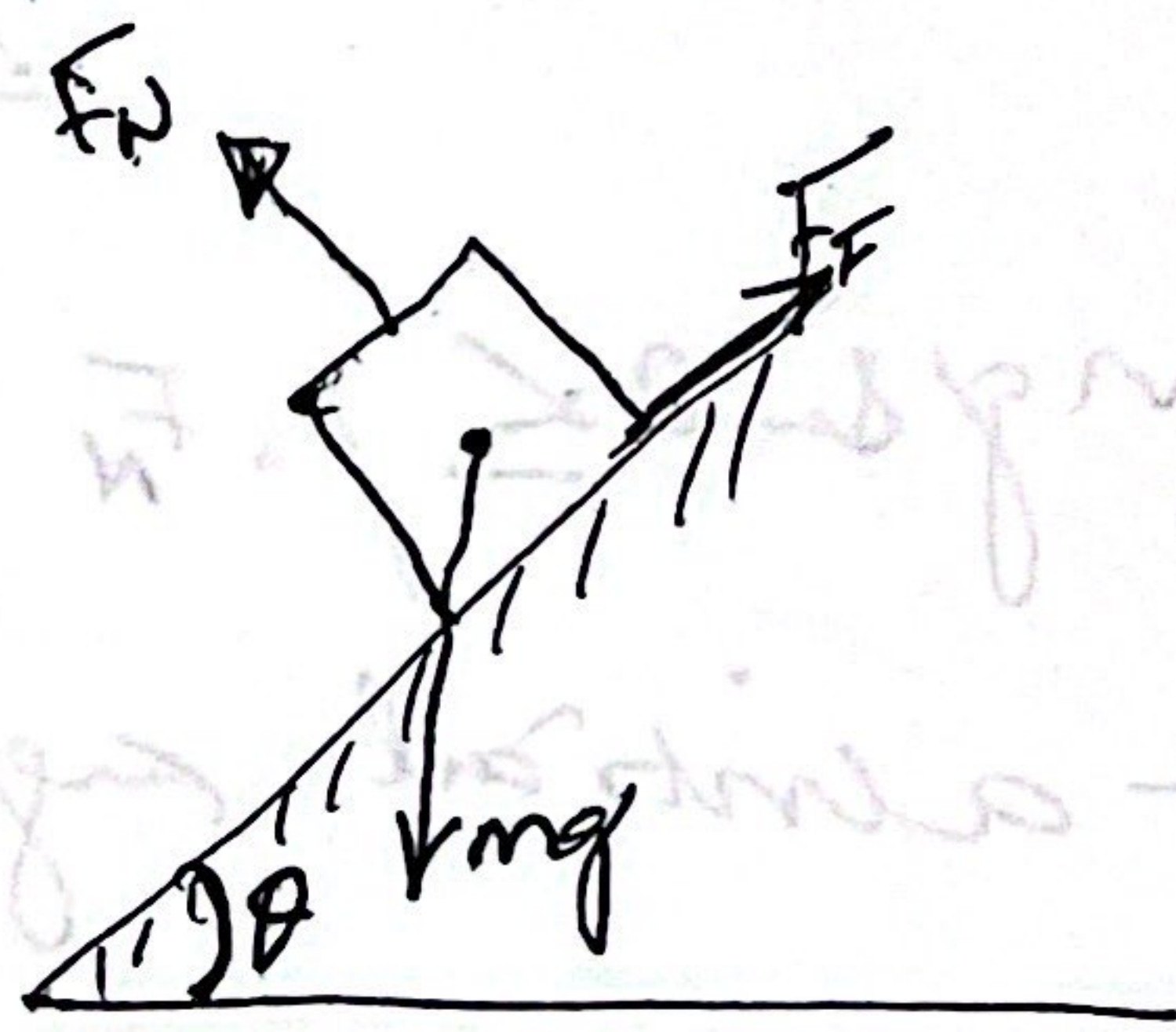
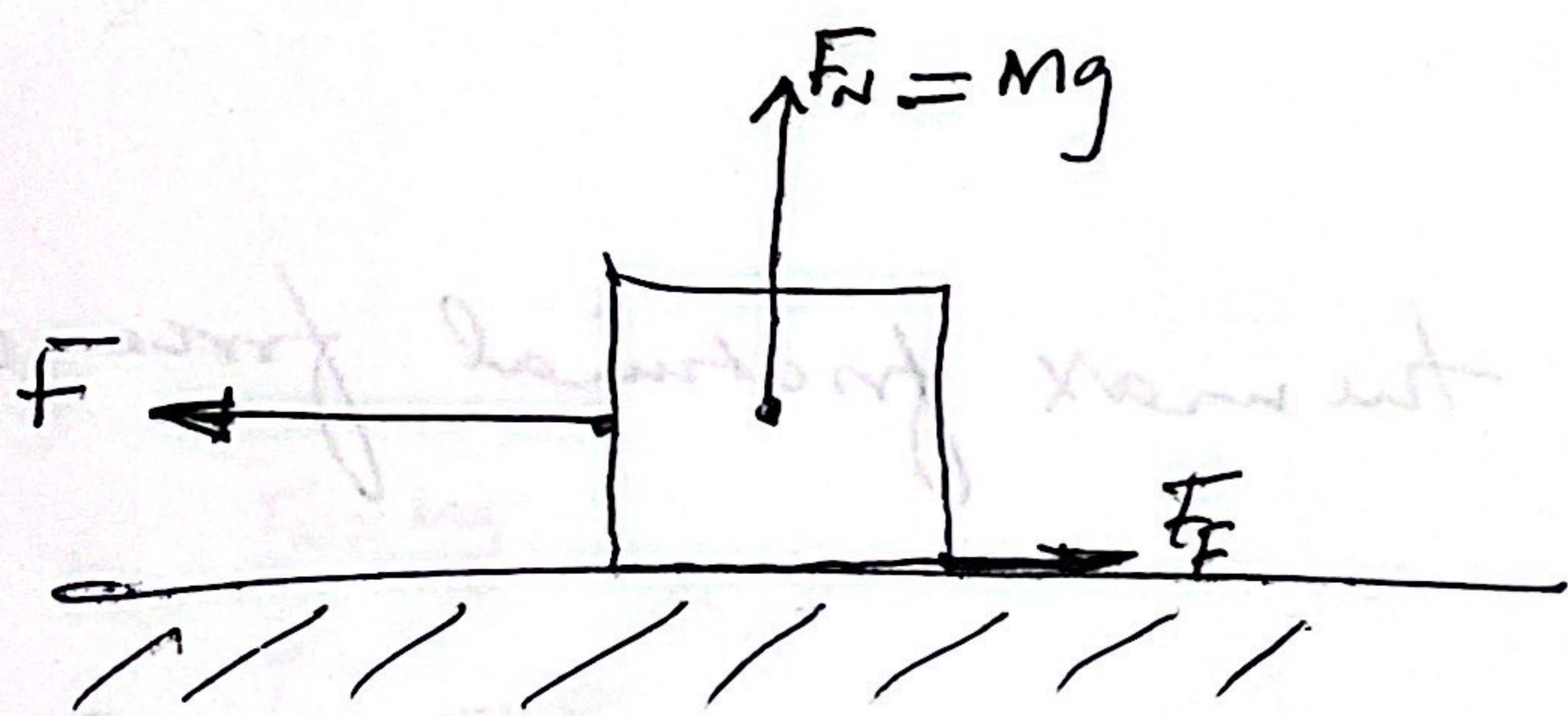
$$F = C V^2$$

C depends on fluid density and the area of the object
 V is speed.

[Jets planes fly at > 30000 feet where air density is less.]

Due to the complexity of materials, shape, size, ^{nature of} lubricating agent present it is not possible to express frictional forces in detailed mathematical form. But generally, the for many surfaces, the sliding frictional forces increase with the force pressing the surfaces together all other things being equal.

Let the force acting to two surfaces be labelled F_N , the normal force. Then the force of friction F_f acting to prevent the motion of sliding is $\leq \mu F_N$



μ applies to surfaces at rest with respect to each other, it is called the coefficient of static friction. When the object starts moving, the coefficient of sliding (dynamic or kinetic) friction is used.

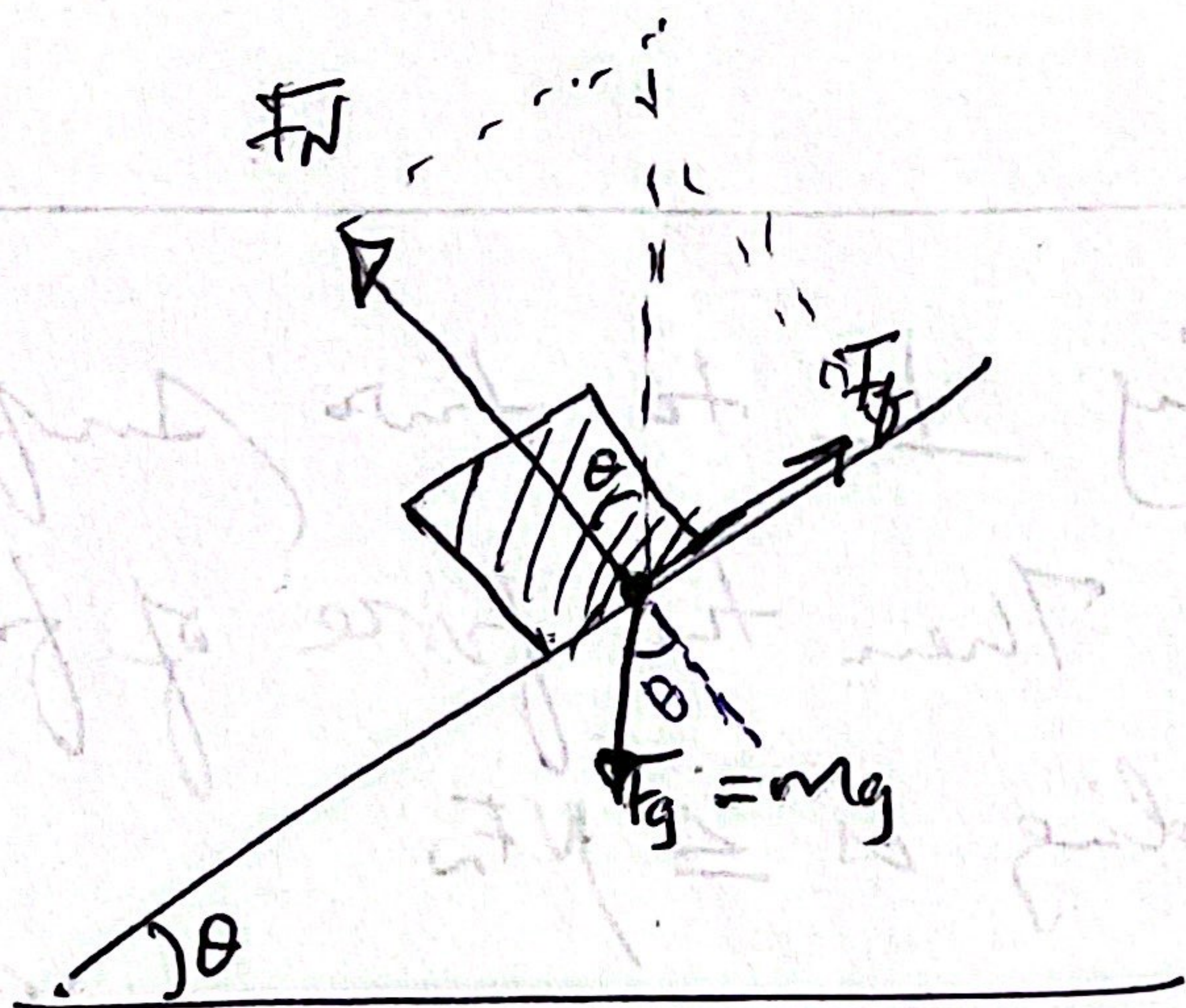
For static contact, the coefficient of static friction μ_s is defined as $\mu_s = \frac{F_f}{F_N}$.

For sliding contact, the coefficient of kinetic friction μ_k is defined as $\mu_k = \frac{F_f}{F_N}$.

For dry surfaces, values of μ_s lie typically in the range 0.2 to 1. For sliding objects, $\mu_k < \mu_s$

In the case of ~~a~~ inclined surfaces, there exists an angle θ called critical angle ~~where~~ maximum force of friction is achieved and the object can slide down the plane.

$$\mu_s = \tan \theta$$



$$mg \sin \theta \leq \mu_s F_N$$

at a critical angle θ_c , the max frictional force is

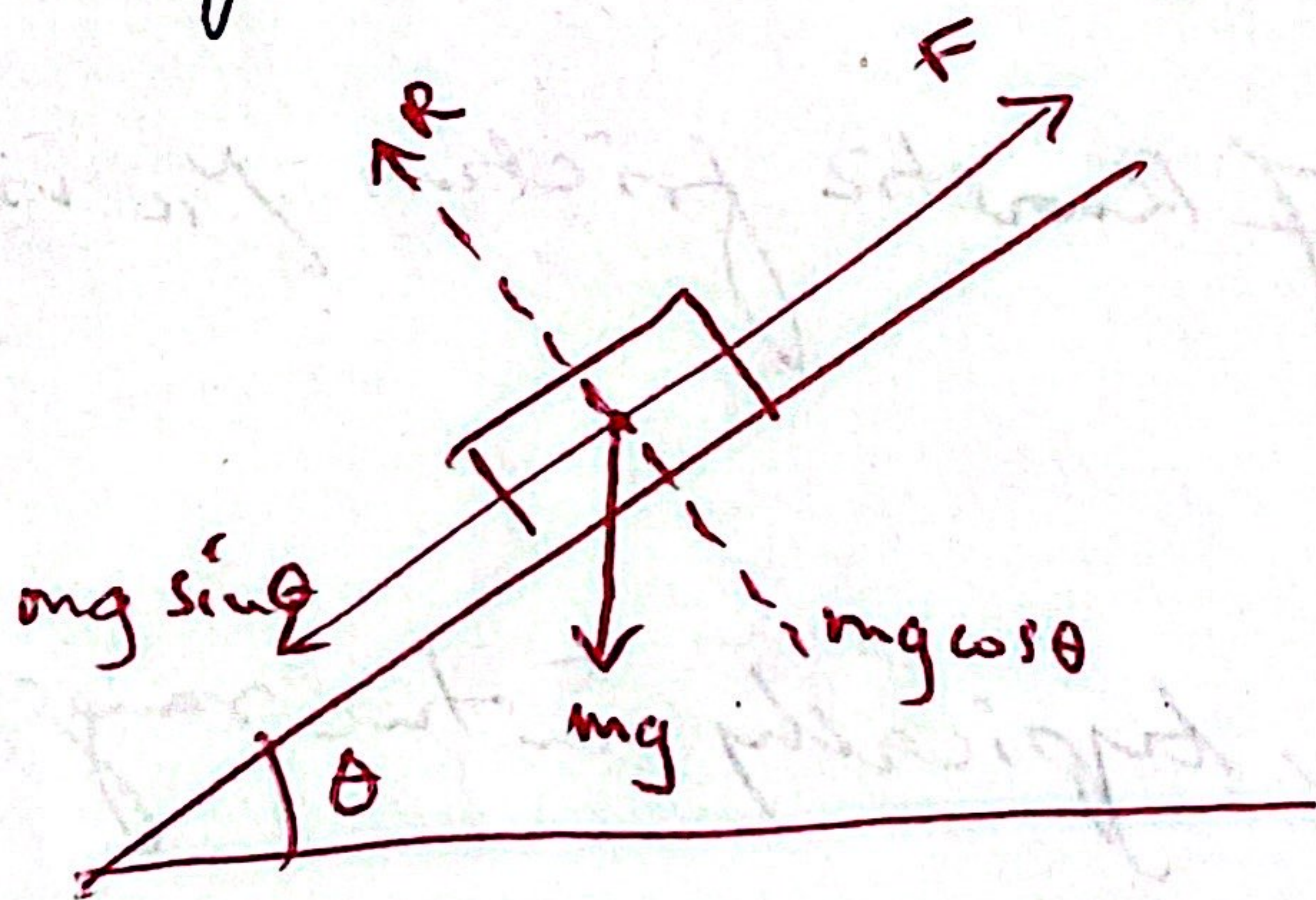
$$mg \sin \theta_c = \mu_s F_N$$

At this angle $F_N = mg \cos \theta_c$

$$\therefore \sin \theta_c = \mu_s \cos \theta_c$$

$$\tan \theta_c = \mu_s$$

Beyond θ_c sliding takes place.



$$F = mg \sin \theta$$

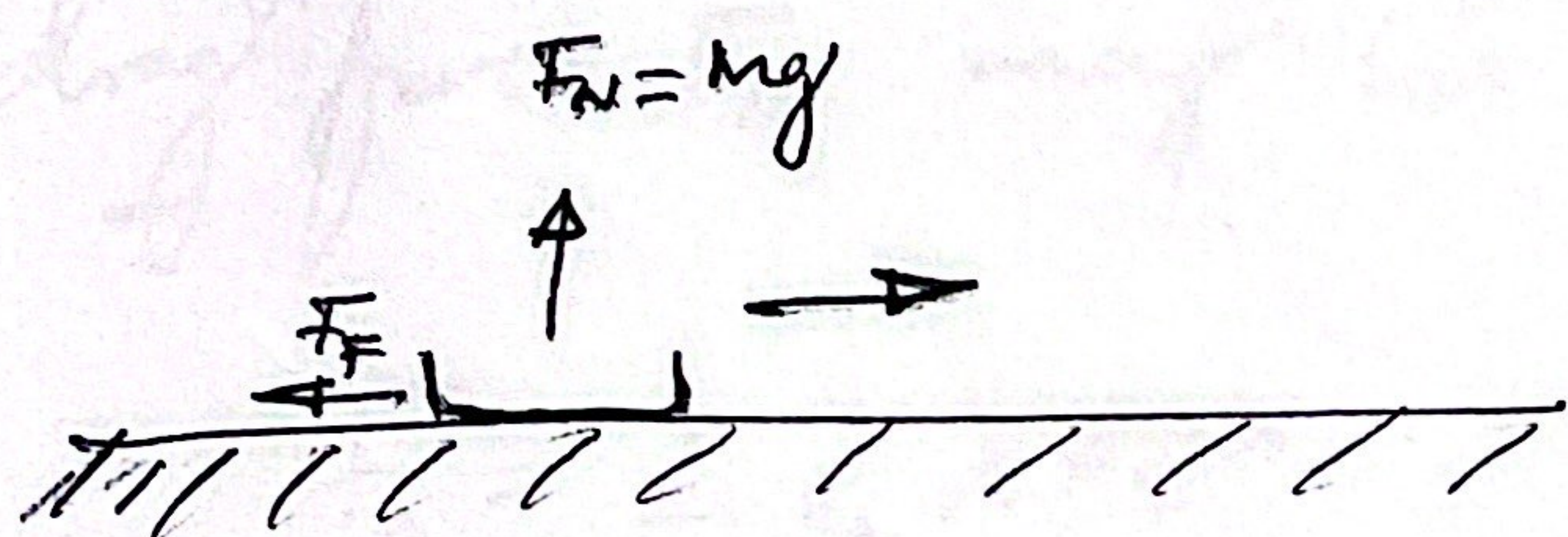
$$R = mg \cos \theta$$

$$\mu = \frac{mg \sin \theta}{mg \cos \theta} = \tan \theta$$

Example

A plate is sent sliding across a counter with an initial speed of 2.5 m/s . If $\mu_k = 0.3$ how far does the plate slide?

S/.



The normal force on the plate = weight mg . The horizontal frictional force is therefore $F_f = \mu_k mg$

The acceleration of the plate has magnitude $a = \frac{F_f}{m}$ or

$$a = \frac{\mu_k mg}{m} = \mu_k g.$$

Since this a is opposite to horizontal forward velocity, the appropriate eqn to use is

$$V^2 = V_0^2 - 2as$$

Final $V=0$ $s = \frac{V_0^2}{2a} = \frac{V_0^2}{2\mu_k g} = \frac{(2.5 \text{ m/s})^2}{2 \times 0.3 \times 9.8 \text{ m/s}^2} = 1.06 \text{ m}$

Any fluid exerts a frictional drag on an object moving through it. For small objects moving slowly, (grain of dust sinking in water,) the frictional force is proportional to the speed of the object.

$$F \propto v$$

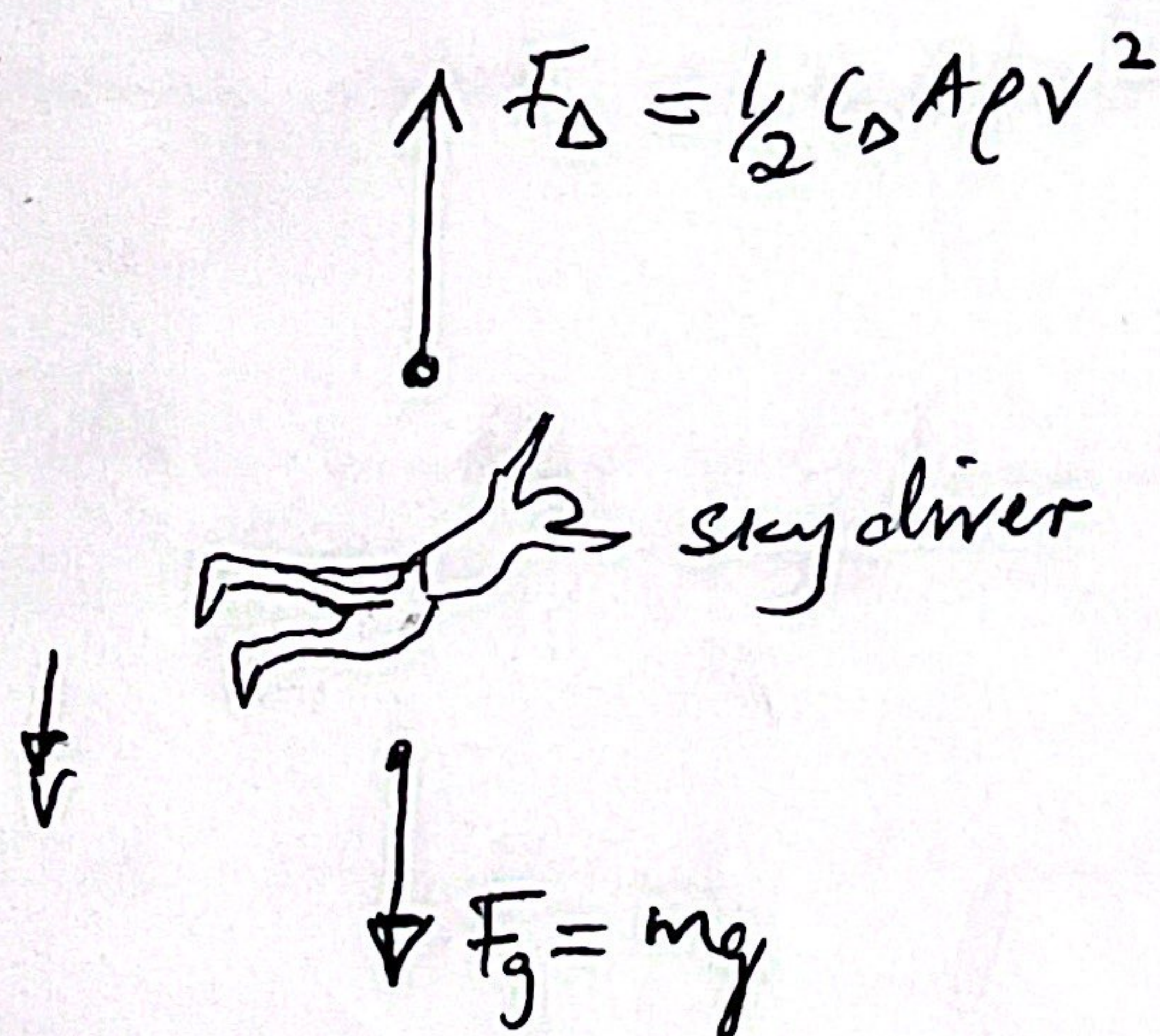
For large objects moving fast, the F varies with v^2 .

The drag force F_D may be written

$$F_D = \frac{1}{2} C_D A \rho v^2$$

C_D is the drag coefficient, A = area, ρ = density, v = speed.

For particular shapes (spheres) the drag coefficient is nearly a constant over a rather wide range of sizes and speeds.



When there is no net acceleration the object reaches a limiting velocity called terminal velocity. It is given by

$$0 = mg - \frac{1}{2} C_D A \rho v^2 \quad (\text{drag force} = \text{gravitational force})$$

Example.

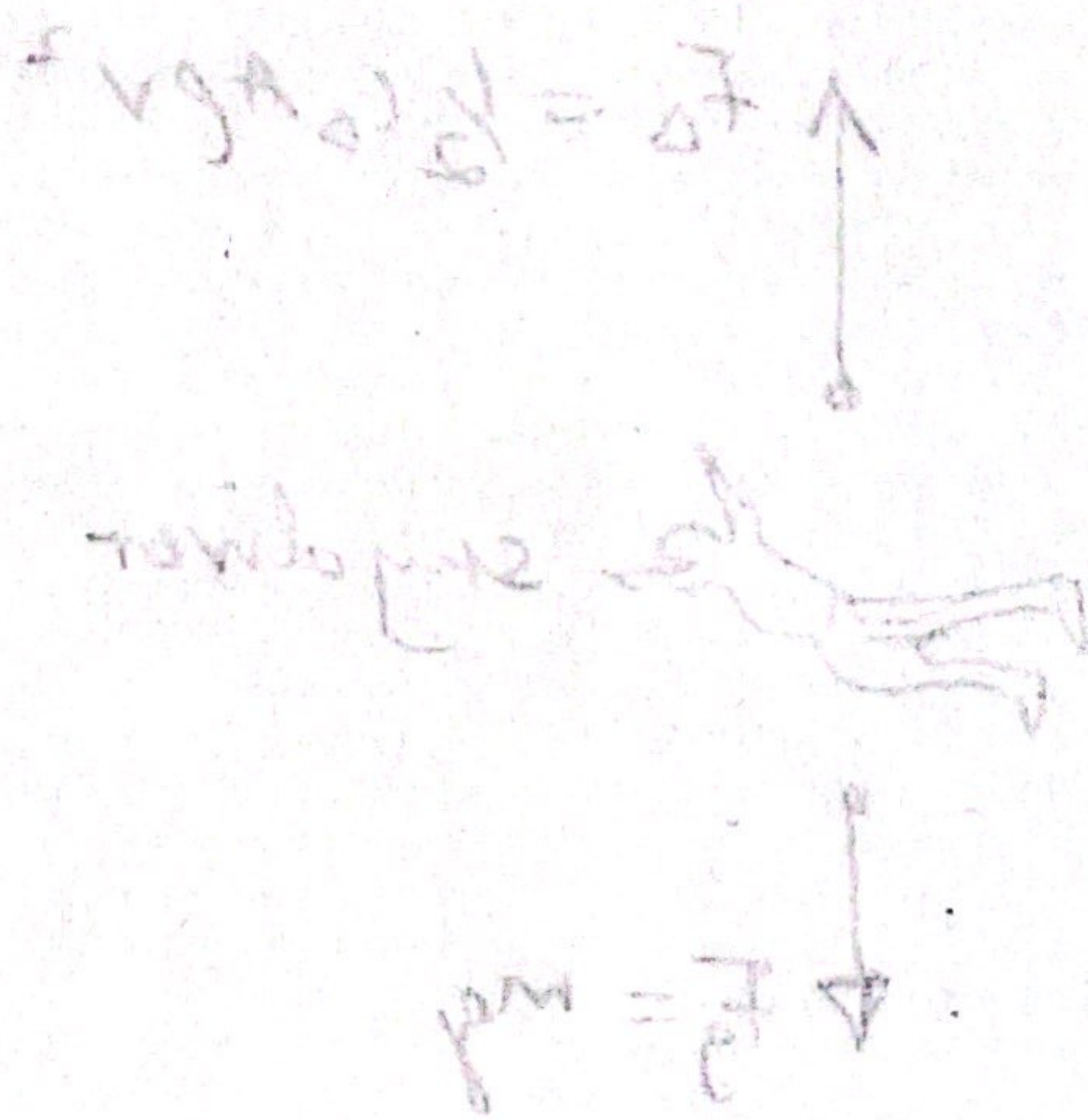
What is the terminal speed in air of a ball of mass 2.5 gm and cross sectional area 11.1 cm^2 ? $C_D = 0.45$, $\rho_{\text{air}} = 1.20 \times 10^{-3} \text{ gm/cm}^3$.

Sol.

$$V_T = \sqrt{\frac{(2)(2.5 \text{ gm})(980 \text{ cm/s}^2)}{(0.45)(11.1 \text{ cm}^2)(1.20 \times 10^{-3} \text{ gm/cm}^3)}} = \sqrt{81.7 \times 10^4 \text{ cm}^2/\text{sec}^2} = 904 \text{ cm/sec}$$

$$= 9.04 \text{ m/s}$$

□ Calculate roughly the terminal speed of a sky diver in air near sea level. Make reasonable estimates of his (or her) mass and cross sectional area; use $C_D = 0.5$.



$$\text{or } V_T = \sqrt{\frac{2mg}{C_D A \rho}}$$

i.e. drag force = gravitational force.

For a tiny ~~sphere~~ ^{object} moving slowly through a fluid; the law of drag force is given by

$$F_D = 6\pi\mu r v$$

This formula is called Stokes law. States that the drag force is directly proportional to the radius r of the sphere and to its speed v .

μ is called the viscosity of the fluid (kg/ms).

Viscosity is a measure of the shear stress that exist in a fluid or a measure of the frictional force acting on a fluid flowing over a solid surface.

The Reynolds number defined by $R = \frac{\rho v d}{\mu}$
 ρ = density, v = velocity, d = diameter, μ = viscosity

determines the different regimes of flow in which different laws of drag are applicable.

R	law of drag force
0 to 1	Stokes
1 to 300	transition region
300 - 300000	$F_D \propto v^2$
> 300000	roughly "

Exercises (Tut 2)

8 (Q 1).

- A pellet of $\phi 2 \times 10^{-4} \text{ m}$ and $m = 10^{-8} \text{ kg}$ falls thru' water of $\mu = 1.0 \times 10^{-3} \text{ kg/ms}$. Assuming validity of Stokes law, calculate the terminal speed of the pellet. Find its Reynolds no at its terminal speed. Estimate the time the pellet needs to reach V_T if it starts from rest.
- Two spheres whose radii differ by a factor of 2 are observed to remain together as they fall, both of them reaching same terminal speed. What is the relative density of the two spheres?
- The $\mu = 1.8 \times 10^{-5} \text{ kg/ms}$ for air. What range of speeds a ball of $\phi 2 \text{ cm}$ is the force of air drag given by Stokes law? For what range of v is the drag force proportional to v^2 ?
- The viscosity of air is $\mu = 1.8 \times 10^{-5} \text{ kg/ms}$. For what range of speed of a ball of diameter 2 cm is the force of air drag given by Stokes' law. For what range of speed is the drag force proportional to v^2 ? [use Reynolds nos in notes]